



Koneru Lakshmaiah Education Foundation

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OPERATING SYSTEMS AND SYSTEMS PROGRAMMING (25CS2104E)

2026–27, Term-I

Project Problem Statement Submission Form

Section No.: -02

Team No.: -20

Project Title:- Linux Debugging System

Team Members

Roll Number	Student Name	Signature
2520030188	U.PRANEETH	
2520030573	M. HARSHITH	
2520030034	U.PRATHIK	

1. Abstract:-

The **Linux Debugging System** is a Linux-based systems programming project focused on debugging and analyzing program execution in a Linux environment. The project demonstrates Operating Systems and system-programming concepts through Linux/POSIX mechanisms and debugging/analysis tools.

The system is intended to support the study and analysis of program behavior, processes, system calls, signals, file operations, memory management, and other relevant Linux mechanisms. The implementation will be developed using C/C++ on Ubuntu/Linux and may make use of appropriate debugging and analysis tools such as GDB and strace. The project will include system design, working implementation, testing, experimental results, and behavior or performance analysis. The expected outcome is a functional Linux-based debugging system that demonstrates practical Operating Systems and Systems Programming concepts and can be presented through a final working demonstration and viva.

2. Problem Statement:-

Debugging programs in a Linux environment requires understanding both program execution and Operating Systems mechanisms. Developers need to analyze

program behavior and identify problems using appropriate system-programming and debugging mechanisms. This project addresses the need for a Linux-based debugging system that demonstrates practical debugging and analysis using Linux/POSIX concepts and tools.

3.Objectives:-

A. To develop a Linux-based debugging system using C/C++.

B. To demonstrate relevant Operating Systems and Linux/POSIX system-programming concepts.

C.To analyze program behavior using appropriate Linux debugging and analysis mechanisms.

D. To test the implementation and evaluate its behavior and performance.

4. Proposed Methodology:-

The project will be designed and implemented as a **user-space Linux/system-programming application**. The implementation will use C/C++ in an Ubuntu/Linux environment and appropriate Linux/POSIX mechanisms. Relevant debugging and analysis tools such as GDB and strace will be used where appropriate. The system will be designed, implemented, tested using test cases, and evaluated through experimental results and behavior/performance analysis.

5. Operating Systems Concepts / Linux APIs Used:-

OS Concept / Linux API / System Call	Purpose in the Project
PROCESS CREATION AND MANAGEMENT	PROGRAM/PROCESS EXECUTION AND ANALYSIS
LINUX SYSTEM CALLS	SYSTEM-LEVEL INTERACTION
FILE I/O	FILE – RELATED OPERATIONS AND ANALYSIS
GDB	DEBUGGING AND ANALYSIS
STRACE	SYSTEM-CALL TRACING AND ANALYSIS
POSIX MECHANISMS	LINUX SYSTEM PROGRAMMING

6. Individual Contribution

Roll Number	Student Name	Individual Responsibility
2520030188	U.PRANEETH	PROJECT DESIGN AND PROCESS/SYSTEM – PROGRAMMING IMPLEMENTATION
2520030573	M.HARSHITH	LINUX DEBUUGING AND ANALYSIS IMPLEMENTATION
2520030034	U.PRATHIK	TESTING AND EXPERIMENTAL IMPLEMENTATION

7. Tools / Platforms / Software Used:-

(List the programming languages, operating system, libraries, tools, and software planned for the project.)

Tool / Platform / Software	Purpose
Linux / Ubuntu	Implementation and demonstration platform
C / C++	Project implementation
GDB	ANALYSIS
STRACE	SYSTEM-CALL ANALYSIS
GITHUB	SOURCE – CODE REPOSITORY

8. Expected Outcome:-

The expected outcome is a working Linux-based Debugging System implemented as a user-space systems-programming application. The project should demonstrate relevant Operating Systems

and Linux/POSIX concepts, provide working implementation evidence, produce test results, and support performance or behavior analysis.

9. GitHub Repository:-

GitHub Repository Link:-

https://github.com/uppusonu/2520030188_573_34---OSSP

Repository Created: - YES

Faculty Name and Signature

Faculty Name:- K. ANUSHA MAM

Signature: _____

Remarks: _____

Faculty Use

Project Approved: _____

☐ Yes ☐ No

Date: _____

Faculty Signature: _____